
LECTURE NOTE

Lecture 4 & 5: Networks - Application Layer

AI510 - Cybersecurity and Innovation

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1 Creating Network Applications

Networks are made on a lower layer such that when creating application, we don't have to create software for the network.

- The server (host) is always available, with a permanent ip address
- clients contact the host
- may be intermediate
- ip may be dynamic

1.1 Peer to peer architecture

Alternative server method, which is not always on.

- Systems instead communicate directly

1.2 Process communicating

- The host has a program that is running
 - Within the host there is inter-process communication (OS)

1.3 Sockets

Interface to the lower layers within a system. To receive message we need to identify ourselves, using an IP **address** and a **port**

2 Application layer protocols

- Types of messages received: request and responses
- Message syntax
- Message semantics
- Rules
- Open protocols
- Proprietary protocols

2.1 What it needs to provide

- **Data integrity:**
 - reliable data transfer, though some specific apps might tolerate some data loss (audio)
- **Timing:**
 - low delay (like video/call chat or games)
- **Throughput:**
 - Consistent throughput (streaming)

3 HTTP - hypertext transfer protocol

- HTTP uses TCP to exchange messages between browser (HTTP client) to server (HTTP host)
 - This is “stateless” in the sense that it doesn't keep information about past clients.

3.1 Non-persistent HTTP

1. client initiates connection on port 80
 2. server accepts
 3. client requests files
 4. server closes TCP connection
 5. client receives message (html file)
- Round-trip time (RTT)
 - requires 2 RTT per. object 🤔

Status line (status code)

3.2 Persistent HTTP

is better

3.3 HTTP messages

Theres two types of HTTP messages: request and response

- requests
 - GET: sends data to server in URL using ? and &
 - POST: sends data to server in message body. (eg. form input)
 - HEAD: requests header (only) that would be returned if specified URL were requested with a HTTP GET method
- responses
 - status codes:
 - 200 OK
 - 301 Moved Permanently
 - 302 Moved Temp
 - 400 Bad request
 - 403 Forbidden
 - 404 Not found
 - 500 Error
 - 418 ... 🤖

3.4 Cookies

Cookies help maintaining state from HTTP (like remembering a shopping cart on a webshop) They can be used for:

- Authorization
- Shopping carts
- Recommendations
- User session states (web e-mail)

Cookies are sent via http, which means that the website does not need to keep your “state”.

Note: cookies permit sites to learn a lot about you on their site

3.4.1 Cookies for ads

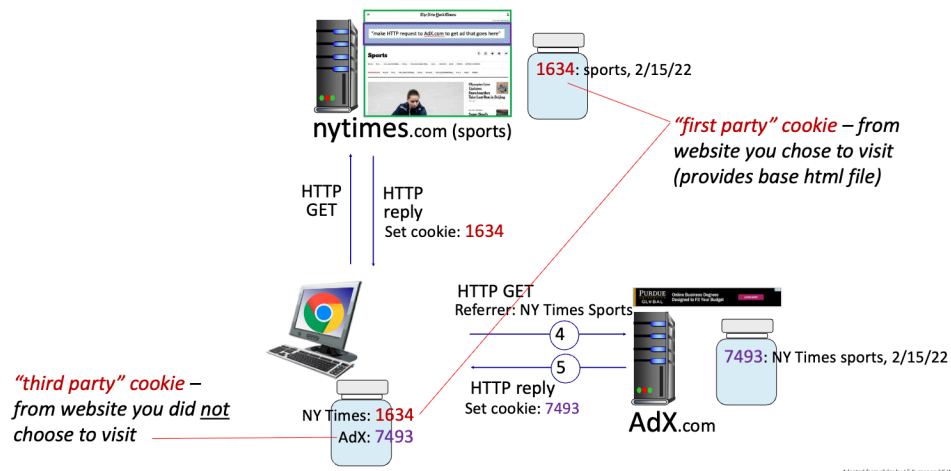


Figure 1:

3.4.2 Caches in HTTP performance

Instead of everyone at your place accessing the same origin server. One can just use caches on the home server.

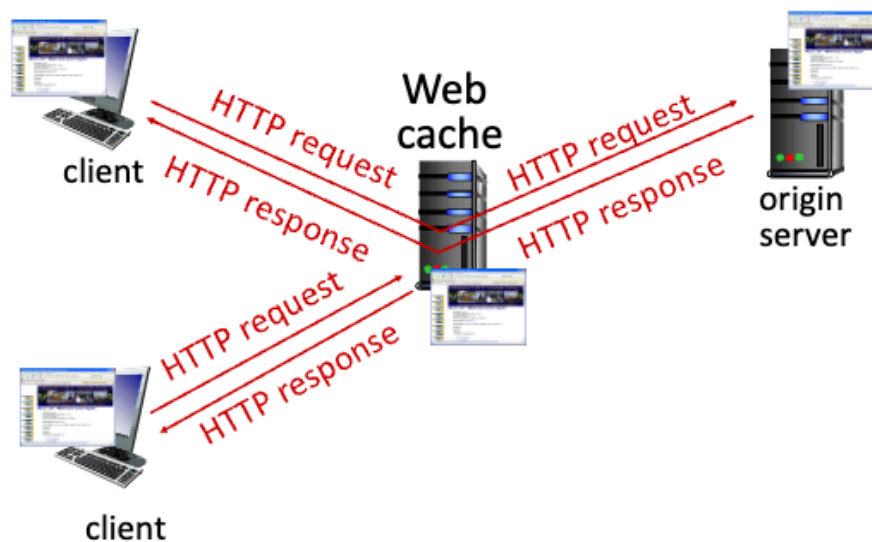


Figure 2:

In the same sense. If you have just seen a website, just save the resources in a cache and load them next time you visit (if its unmodified).

3.5 Email protocols

Mail serves only have: A mailbox and a message queue

3.5.1 Simple Mail Transfer Protocol

Uses **TCP** to reliably transfer email message from client.

```
S: 220 hamburger.edu
C: HELO crepes.fr
S: 250 Hello crepes.fr, pleased to meet you
C: MAIL FROM: <alice@crepes.fr>
S: 250 alice@crepes.fr... Sender ok
C: RCPT TO: <bob@hamburger.edu>
S: 250 bob@hamburger.edu ... Recipient ok
C: DATA
S: 354 Enter mail, end with "." on a line by itself
C: Do you like ketchup?
C: How about pickles?
C: .
S: 250 Message accepted for delivery
C: QUIT
S: 221 hamburger.edu closing connection
```

Figure 3:

*Note: This is only for **sending** mails*

3.6 SMTP vs HTTP

- HTTP
- Client Pull
- ASCII commands
- ASCII status codes
- Each object encapsulated in its own response message
- Non-persistent & persistent connections
 - SMTP
- Client Push
- ASCII commands
- ASCII status codes
- Multiple objects sent in multipart message
- Persistent connections

3.7 Retrieving mails: Mail access protocols

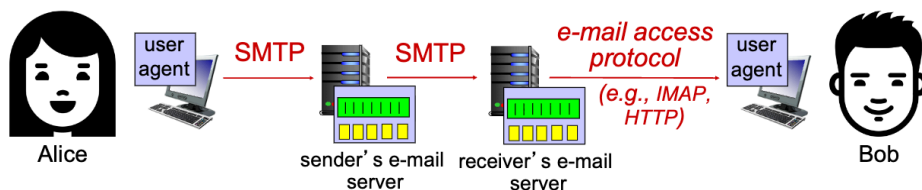


Figure 4: fig

IMAP: Internet Mail Access Protocol [RFC 3501]: messages stored on server, IMAP provides retrieval, deletion, folders of stored messages on server

or use POP

4 Securing E-mail

- Confidentiality of message content
- Authenticity of sender

4.1 Option 1: Transport Layer Security

Assumptions:

- Clients and servers trusted
- Network untrusted

Protocols encapsulated in secure connections on transport layer (TLS)

- Advantage: easier, since it only needs to be deployed by servers
- Disadvantage: servers must be trusted

4.2 Option 2: End-to-end Security

Assumptions:

- Clients trusted
- Network and servers untrusted

Message body encrypted/decrypted & signed/verified by user agents

- Advantage: Strictly more secure
- Disadvantage: More complicated for users

4.3 OpenPGP

Technically the older standard, due to predecessor PGP (1991)

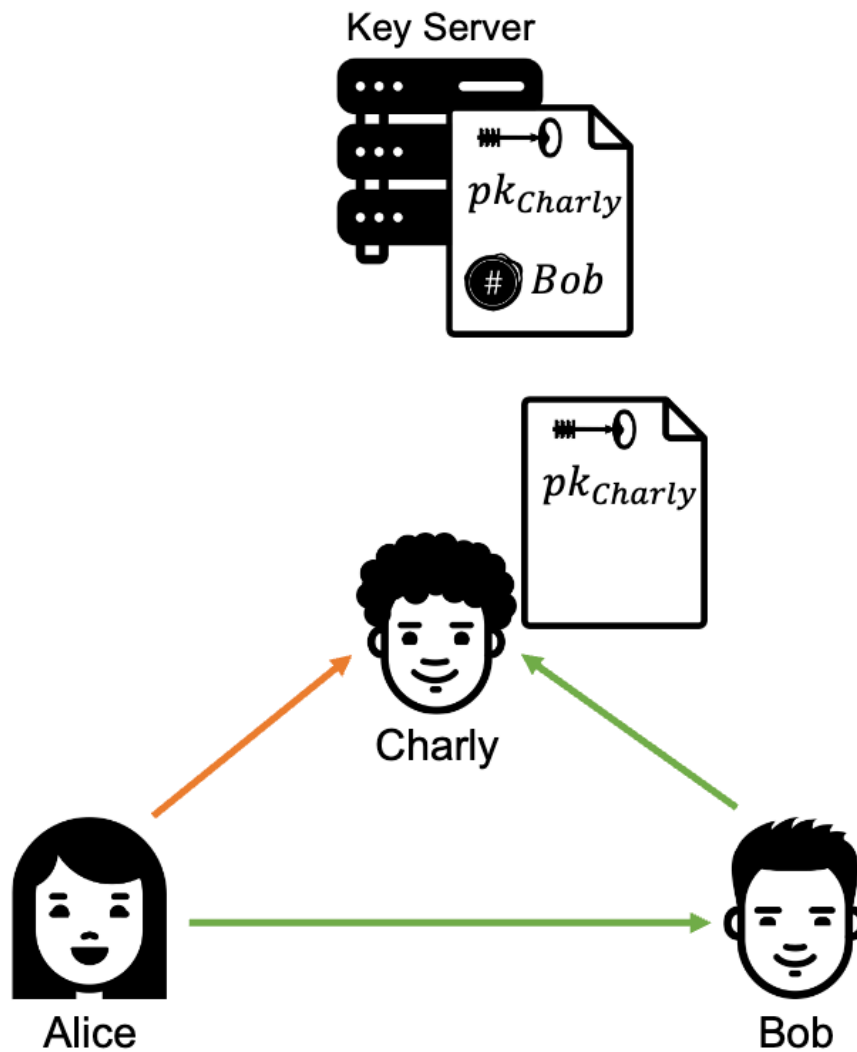


Figure 5:

5 DNS

Allows you to find the right host to contact. (you might have some info on someone, but not their ip address)

- DNS is a distributed database to map between IP address and name, and vice versa
- DNS is a hierarchical database

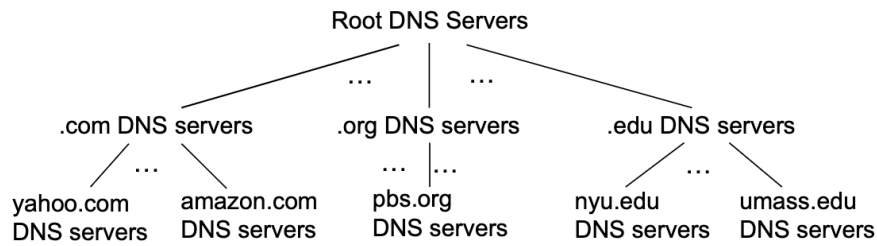


Figure 6:

DNS information can also be cached

5.1 Records

- type=A/AAAA
 - name is hostname
 - value is ip address
- type=NS
 - name is domain (like cogito.dk)
 - value is hostname of authoritative name server for this domain
- type=CNAME
 - name is s alias name for some “canonical” (the real) name
 - value is canonical name
- type=MX
 - value is name of SMTP mail server associated with name

5.2 Security

- DDos attacks by redirecting long requests to a spoofed IP
- Spoofing attacks, intercept DNS queries.

5.3 DNS against phishing

5.3.1 Spoofing

Assume email sender spoofing like shown below

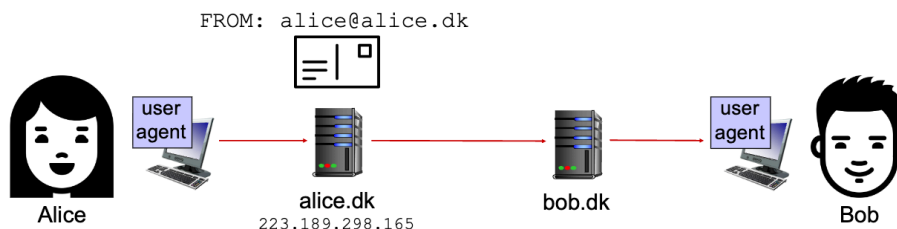


Figure 7: Alice might not be who she is saying she it.

We can use dns to confirm ip is correct.

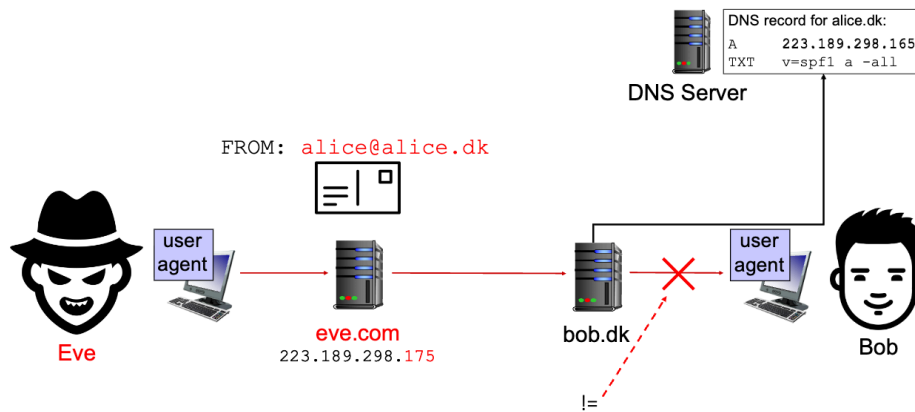


Figure 8: Spoofer is caught using DNS

5.3.2 Message Manipulation

Assume that someone might intercept your message, and corrupt it

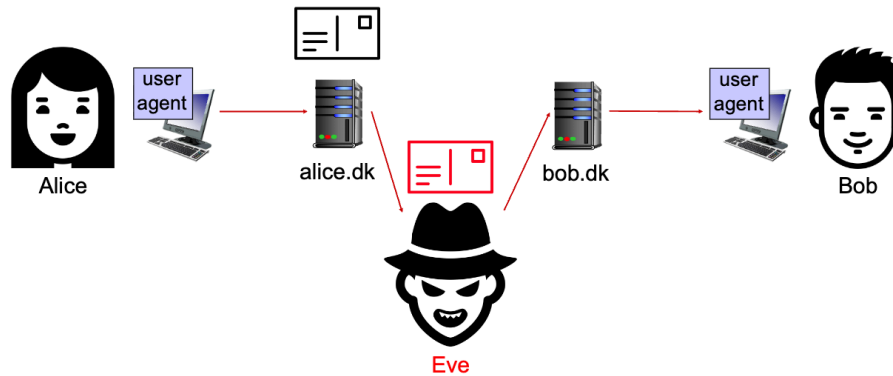


Figure 9:

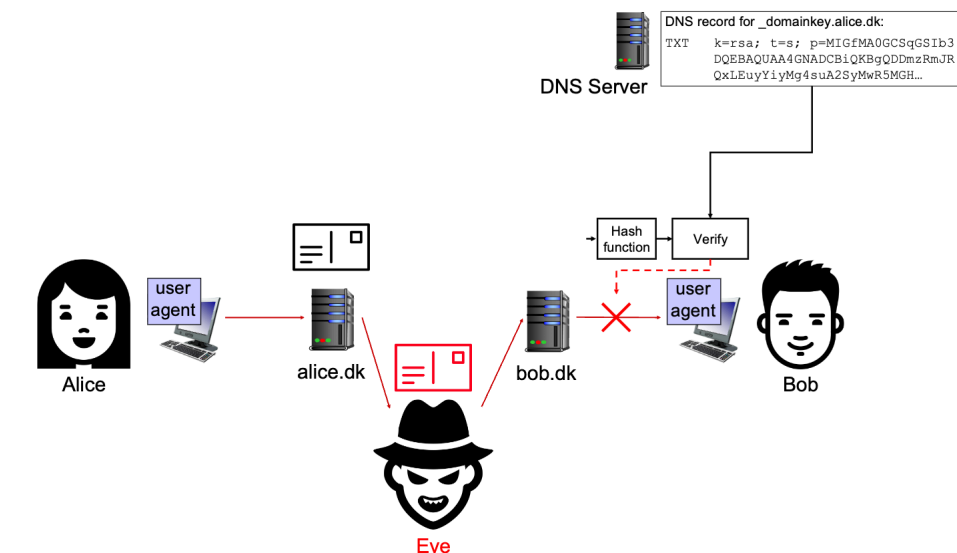


Figure 10:

6 Next Lecture

- TCP (transmission control protocol)
 - Provides control in transport and flow
 - Does not provide good timing
- UCP
 - Provides good timing and security with little control